

Harava Insights

Client Financial Dashboard - Frontend Integration Guide

How to consume the dashboard insight endpoints, the response shapes, and the recommended chart design for each widget. All figures are computed live from each company's mirrored QuickBooks data.

1. Overview & authentication

The dashboard is powered by **ten section endpoints**. Each returns a single, pre-aggregated JSON payload for one dashboard area, so the frontend fetches a section and renders it directly - no client-side aggregation.

There are two parallel sets of the same endpoints:

- **Firm console** (owner/admin): `/api/v1/tenant/companies/{companyId}/dashboard/...` - pass any company in the tenant.
- **Client portal** (company user): `/api/v1/portal/dashboard/...` - company is taken from the auth token, never the URL.

Both require a Bearer token: `Authorization: Bearer <token>` (firm uses the staff token; portal uses the CLIENT token). Responses are identical in shape. A missing/expired token returns **401**; a company outside your scope returns **404**.

2. Conventions

- **Currency**: every payload includes a currency ISO code (e.g. USD, NGN). Amounts are plain numbers in that currency - format client-side.
- **Percentages**: fields ending in Pct are 0-100 (e.g. 41.0 = 41%).
- **months param**: trend endpoints accept `?months=N` (default 12, max 36). Periods are "YYYY-MM".
- **Empty is normal**: a company that doesn't use a feature (no inventory, no classes) returns empty arrays / zeros. Hide or grey those widgets - see §5.
- **Caching**: data refreshes on the sync cadence (~15 min), so caching a section for a few minutes client-side is fine.

3. Endpoints & widget designs

Path shown is the firm-console form; the portal form is the same with `/portal/dashboard` and no `companyId`.

3.1 KPI cards

GET `/dashboard/kpis`

```
{
  "currency": "USD",
  "cashOnHand": 92450.10,
  "revenueThisMonth": 18200.00, "revenuePrevMonth": 16400.00, "revenueYtd": 142300.00,
  "expensesThisMonth": 12750.00, "netProfitThisMonth": 5450.00,
  "netMarginPct": 29.9, "grossMarginPct": 41.0,
  "arOutstanding": 23600.00, "arOverdue": 8100.00,
  "apOutstanding": 14200.00, "apDueSoon": 3400.00,
  "activeCustomers": 29, "activeVendors": 26
}
```

Design: One call powers the whole KPI row. Render 6-8 stat cards. Show a delta chip on Revenue (`revenueThisMonth` vs `revenuePrevMonth`), and secondary text on cards (e.g. '\$8.1k overdue' under A/R). Colour net profit green/red by sign.

3.2 Profit & loss

GET /dashboard/pnl?months=12 • param: months

```
{
  "currency": "USD", "income": 142300, "costOfGoodsSold": 84000, "grossProfit": 58300,
  "grossMarginPct": 41.0, "operatingExpenses": 38200, "netProfit": 20100, "netMarginPct": 14.1,
  "trend": [{"period": "2026-02", "revenue": 12100, "expenses": 9800, "net": 2300}, ...],
  "marginTrend": [{"period": "2026-02", "grossMarginPct": 40.2, "netMarginPct": 12.0}, ...]
}
```

Design: Primary chart: combo - bars for revenue & expenses, line for net, over trend[]. Secondary: line chart of marginTrend[] (two series). Optional headline: a P&L waterfall (income -> COGS -> gross -> opex -> net) from the summary totals.

3.3 Cash flow

GET /dashboard/cash-flow?months=12 • param: months

```
{
  "currency": "USD", "cashOnHand": 92450.10, "runwayMonths": 7.2,
  "trend": [{"period": "2026-02", "inflow": 14100, "outflow": 11800, "net": 2300}, ...],
  "balances": [{"id": "35", "name": "Checking", "type": "Bank", "balance": 78200.00}, ...]
}
```

Design: Primary: grouped bars (inflow vs outflow) + net line over trend[]. Side panel: bank balances[] as small bars or cards. runwayMonths -> a gauge or big number (may be null if not burning cash - hide gauge then).

3.4 Balance sheet

GET /dashboard/balance-sheet

```
{
  "currency": "USD", "assets": 210400, "liabilities": 86300, "equity": 124100,
  "assetAccounts": [{"id": "35", "name": "Checking", "amount": 78200}, ...],
  "liabilityAccounts": [...], "equityAccounts": [...]
}
```

Design: Three summary cards (Assets / Liabilities / Equity) + a single stacked bar showing the split. Below, three expandable lists of the top accounts per group (...Accounts[]).

3.5 Receivables

GET /dashboard/receivables

```
{
  "currency": "USD",
  "aging": {"current": 14200, "days1to30": 5200, "days31to60": 2100, "days61to90": 1100, "days90plus": 1000, "total": 23600},
  "totalOutstanding": 23600, "overdue": 9400, "daysSalesOutstanding": 42,
  "paidCount": 21, "openCount": 8, "overdueCount": 5,
  "topDebtors": [{"id": "12", "name": "MTN Ltd", "amount": 6100}, ...]
}
```

Design: Primary: horizontal stacked bar of aging buckets (green->red). KPI: DSO number. Donut: paid/open/overdue counts. Bar: topDebtors[]. A 'chase' CTA on overdue is a nice touch.

3.6 Payables

GET /dashboard/payables

```
{
  "currency": "USD",
  "aging": {"current": 9800, "days1to30": 2400, "days31to60": 1000, "days61to90": 600, "days90plus": 400, "total": 14200},
  "totalOutstanding": 14200, "dueSoon": 3400, "daysPayableOutstanding": 31,
  "topCreditors": [{"id": "7", "name": "Dangote Supplies", "amount": 4200}, ...]
}
```

Design: Mirror of receivables: aging stacked bar, DPO number, 'due this week' KPI (dueSoon), topCreditors[] bar. Pair A/R and A/P side by side on the dashboard.

3.7 Sales

GET /dashboard/sales?months=12 • param: months

```
{
  "currency": "USD",
  "revenueTrend": [{"period": "2026-02", "revenue": 12100, "expenses": null, "net": null}, ...],
  "topCustomers": [{"id": "12", "name": "MTN Ltd", "amount": 31000}, ...],
  "topItems": [{"id": "5", "name": "Consulting", "revenue": 18400, "quantity": 92}, ...],
}
```

```

    "invoiceCount":31,"averageInvoiceValue":1830.50,
    "estimateCount":4,"estimatesConverted":3,"conversionRatePct":75.0,"openPipeline":8200
  }

```

Design: Revenue line (revenueTrend[]). Horizontal bars for topCustomers[] and topItems[]. Stat cards: invoiceCount, averageInvoiceValue. Funnel/number for estimate conversion + openPipeline. Customer-concentration risk = top customer amount / total revenue.

3.8 Expenses

GET /dashboard/expenses?months=12 • param: months

```

{
  "currency":"USD",
  "expenseTrend":[{"period":"2026-02","revenue":null,"expenses":9800,"net":null}, ...],
  "byCategory":[{"id":"60","name":"Payroll","amount":38000}, ...],
  "topVendors":[{"id":"7","name":"Dangote Supplies","amount":21000}, ...],
  "largest":[{"type":"Bill","reference":"1042","party":"Dangote","date":"2026-07-10","amount":38000}, ...]
}

```

Design: Expense bars (expenseTrend[]). Pie or treemap for byCategory[]. Bar for topVendors[]. Table for largest[] transactions.

3.9 Inventory

GET /dashboard/inventory

```

{
  "currency":"USD","totalValue":48200.00,
  "items":[{"id":"5","name":"Widget A","quantityOnHand":120,"unitCost":8.50,"totalValue":1020,"marginPct":46.0}, ...],
  "lowStock":[{"id":"9","name":"Widget X","quantityOnHand":4, ...}]
}

```

Design: Big number: totalValue. Sortable table of items[] (qty, value, margin%). lowStock[] as a flagged list/badge. Hide the whole card if items[] is empty (service company).

3.10 Recent activity

GET /dashboard/activity?limit=20 • param: limit

```

[
  {"type":"Payment","reference":null,"party":"Kola Ent.,"date":"2026-07-18","amount":1200},
  {"type":"Invoice","reference":"1042","party":"Blue Foods","date":"2026-07-17","amount":640}
]

```

Design: A timeline / feed list, newest first. Icon + colour per type (Invoice, Payment, Bill, Sales Receipt, Purchase). Returns a plain array.

3.11 Authoritative QuickBooks reports (separate from the dashboard)

GET /dashboard/reports/{type} • type = any of QuickBooks' 23 reports: profit-loss, balance-sheet, cash-flow, aged-receivables, aged-payables, trial-balance, customer-sales, customer-income, item-sales, vendor-expenses, account-list, aged-receivable-detail, aged-payable-detail, customer-balance, customer-balance-detail, vendor-balance, vendor-balance-detail, class-sales, department-sales, inventory-valuation-summary, profit-loss-detail, general-ledger, transaction-list

```

{
  "reportType": "ProfitAndLoss",
  "generatedAt": "2026-07-22T01:00:00Z",
  "periodStart": "2026-01-01", "periodEnd": "2026-07-22",
  "data": { ...QuickBooks' own report JSON (rows/columns tree)... }
}

```

Design: these are QuickBooks' own reports, so they reconcile to the penny with what the client sees in QuickBooks. The common statements refresh on a schedule; the rest are fetched on-demand the first time they're requested (that call may take a second) and then cached. Keep these in a separate 'Reports' area of the UI, distinct from the interactive dashboard. Render 'data' as a report table, or use it for an 'official statements' view. In addition, the /pnl and /balance-sheet responses above now include an 'authoritative' block (penny-exact totals) whenever a snapshot exists; use it for the headline numbers and keep the computed arrays for the interactive charts. The block is null until the first report refresh, so always fall back to the computed figures.

4. Recommended dashboard layout

Section	Endpoint(s)	Content
Row 1 - KPI strip	kpis	6-8 stat cards across the top.
Row 2 - Money flow	cash-flow + kpis(cash)	Cash in-vs-out combo chart; bank balances beside it.
Row 3 - Profitability	pnl	Revenue/expense/net combo + margin line.
Row 4 - Owed & owing	receivables + payables	A/R aging and A/P aging side by side.
Row 5 - Where it comes/goes	sales + expenses	Top customers & items; expense-by-category.
Row 6 - Context	inventory + activity	Inventory table (if any) + recent activity feed.

5. Empty-state & adaptive rules

Not every business uses every feature. Make widgets adaptive so the dashboard looks tailored for any company:

- **Empty array / zero total** -> hide the widget or show a subtle 'No data yet' tile. Don't render an empty chart.
- **Inventory**: hide the card entirely when items is empty (service businesses).
- **Estimate conversion**: hide when estimateCount is 0.
- **runwayMonths = null**: the business isn't burning cash - show 'Cash-flow positive' instead of a gauge.
- **Classes/Departments**: not exposed as separate endpoints in v1 (many companies don't use them); revisit if a client needs segment reporting.

6. Accuracy notes

Trends (revenue, expenses, cash) are computed from transaction totals: ideal for charts and directionally exact. The balance sheet and cash balances come straight from account balances. Gross margin and COGS use account classification.

For figures that tie out to the penny with QuickBooks' own Profit & Loss, Balance Sheet and Cash Flow, this is now implemented: Harava pulls QuickBooks' own reports from the Reports API and stores them as snapshots (refreshed on a schedule). The P&L and balance-sheet endpoints expose those authoritative totals in an 'authoritative' block, and the full reports are available at /dashboard/reports/{type} (see 3.11).

7. Where AI plugs in later

Every endpoint returns clean numeric aggregates, so the planned AI layer sits on top of these same responses - no rework. Expected additions: revenue & cash-flow **projections**, **anomaly** flags ('expenses up 40% vs usual'), plain-language **summaries**, and **recommendations** ('chase these 3 overdue invoices to recover \$9.4k'). The frontend can surface these as an 'Insights' panel fed by a future /dashboard/ai - * endpoint.

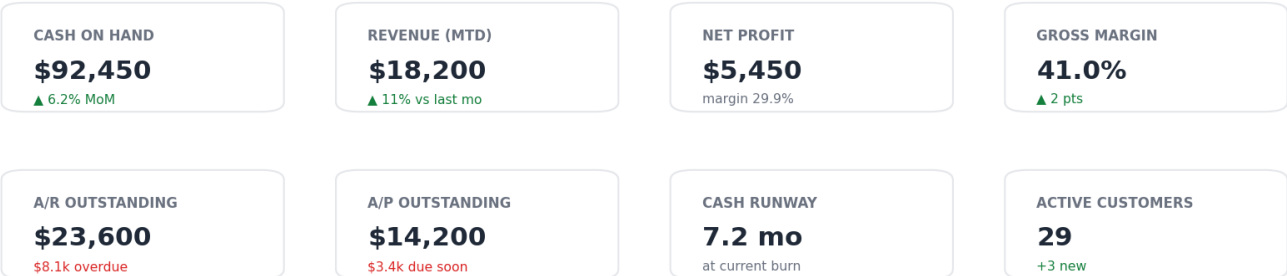
Appendix A - Dashboard visual preview (mockups)

One render per dashboard section, with sample data - covering all ~38 widgets. These show layout, chart types and styling; real figures come from the endpoints above, and widgets should hide gracefully when a company has no data for them (see \$5).

A.1 KPI strip - the 8 headline cards

Headline KPIs (GET /dashboard/kpis)

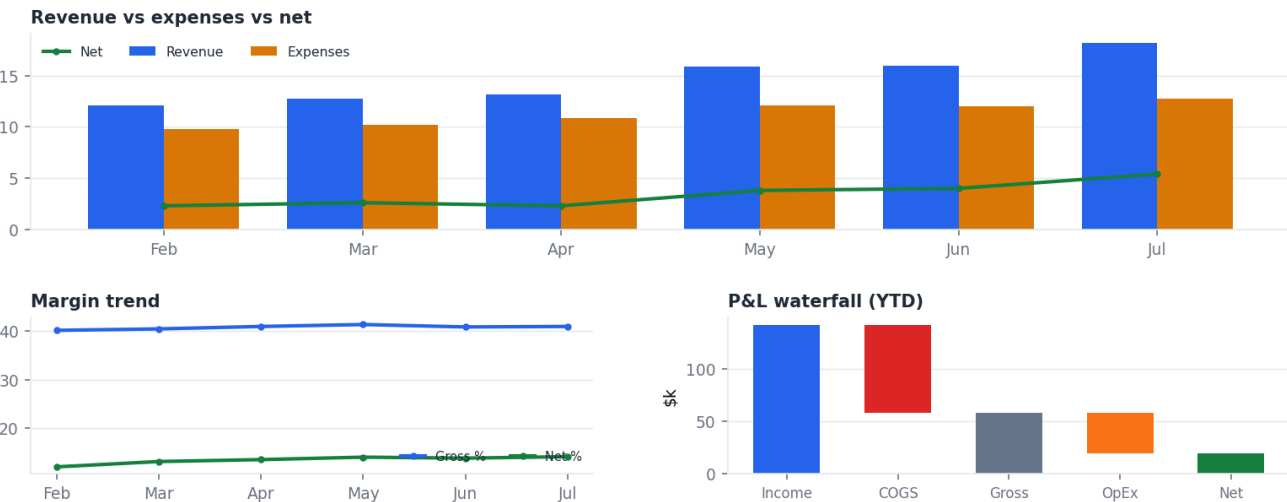
Sample - USD



A.2 Profit & loss - revenue/expense/net, margin trend, P&L waterfall

Profit & loss (GET /dashboard/pnl)

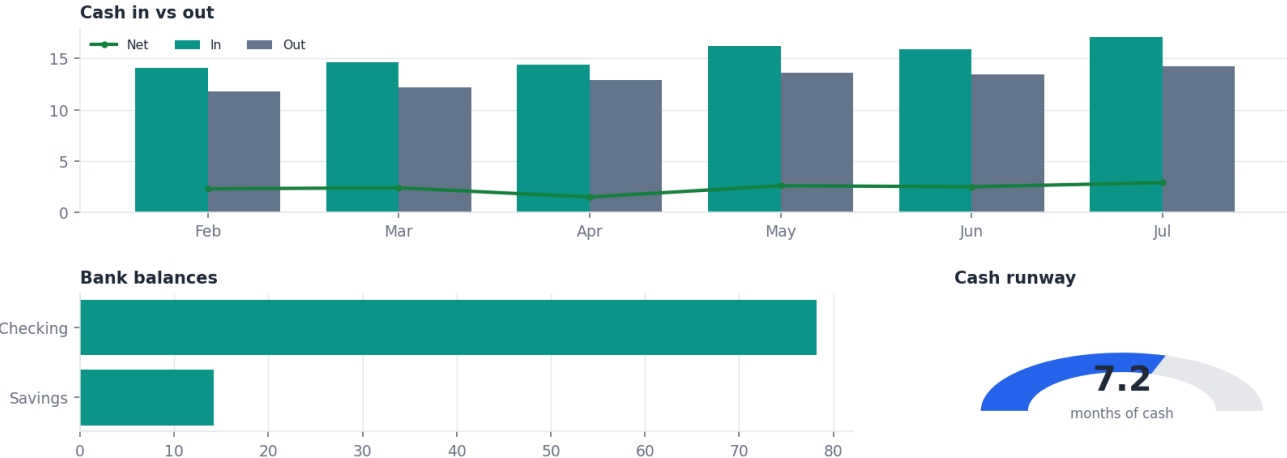
Sample - USD thousands



A.3 Cash flow - in vs out, bank balances, runway gauge

Cash flow (GET /dashboard/cash-flow)

Sample - USD thousands

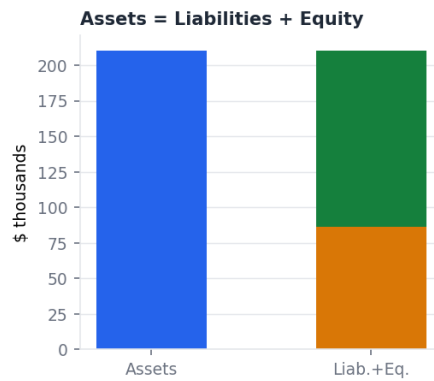


A.4 Balance sheet - assets / liabilities / equity + account breakdown

Balance sheet (GET /dashboard/balance-sheet)

Sample · USD

Assets	\$210,400
Liabilities	\$86,300
Equity	\$124,100



Top asset accounts

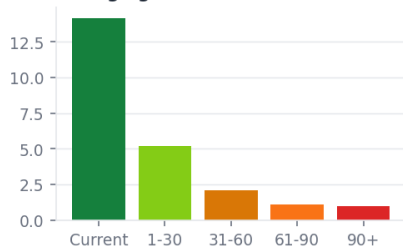
Checking	\$78,200
Accounts Receivable	\$23,600
Inventory	\$48,200
Fixed Assets	\$41,000
Savings	\$14,200

A.5 Receivables & payables - aging, invoice status, DSO/DPO, top debtors & creditors

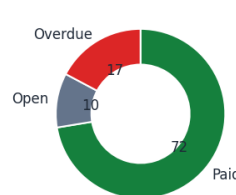
Receivables & payables (GET /dashboard/receivables · /payables)

Sample · USD thousands

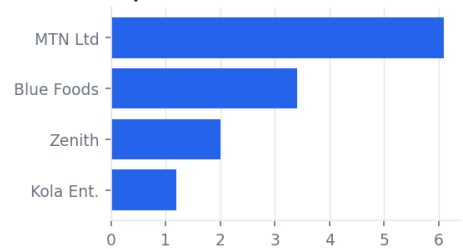
A/R aging · DSO 42d



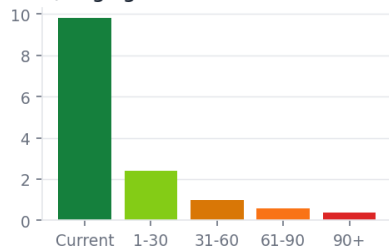
Invoice status



Top debtors



A/P aging · DPO 31d

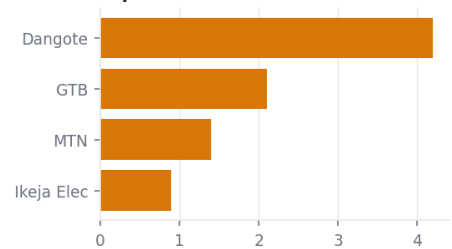


DSO 42 / DPO 31

days to collect / to pay

9.4k overdue · 3.4k due this week

Top creditors



A.6 Sales - revenue trend, top customers & products, concentration, estimate conversion

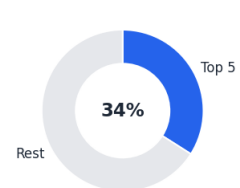
Sales (GET /dashboard/sales)

Revenue over time

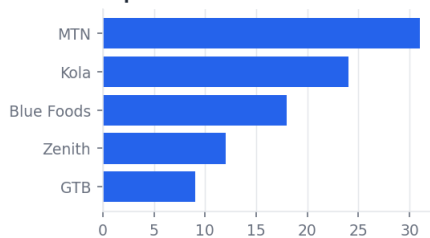


Sample · USD thousands

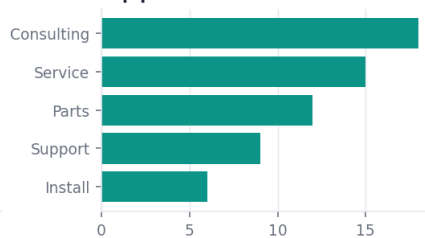
Customer concentration



Top customers



Top products

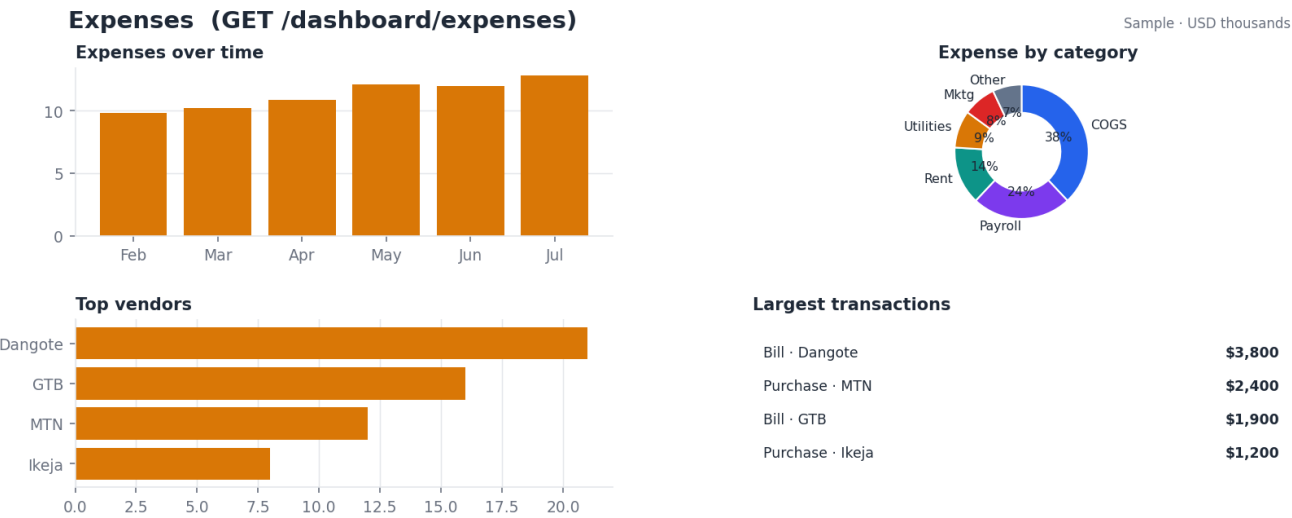


Estimate conversion

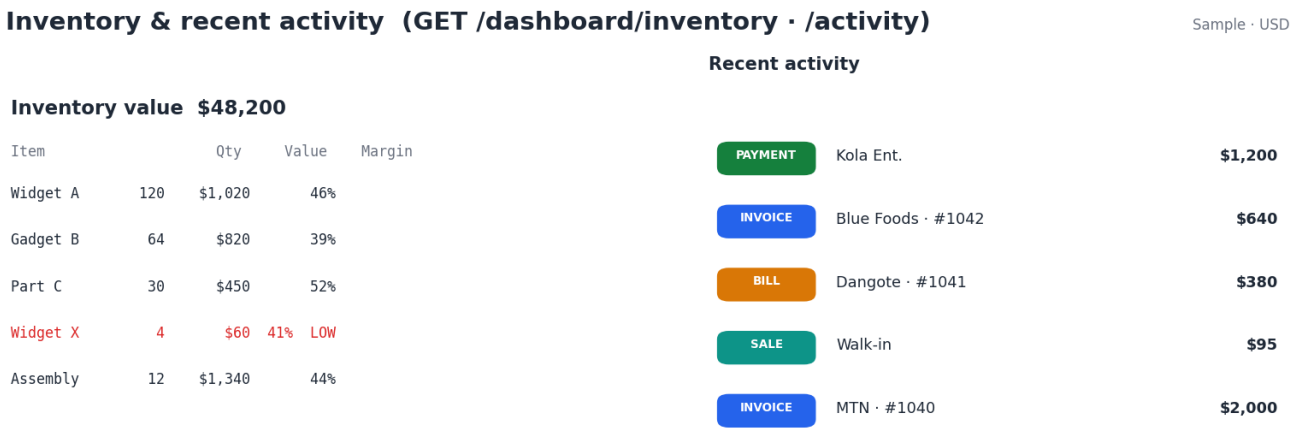


75% · pipeline \$8.2k

A.7 Expenses - trend, by category, top vendors, largest transactions



A.8 Inventory & activity - stock value/margin, low-stock, recent feed



Generated for the Harava frontend team. Endpoints live under the firm console and the client portal; both are covered in the Postman collection (folders 'Client Dashboard (firm console)' and Client Portal -> 'Financial dashboard (my company)').